



A new labdane diterpenoid from *Scoparia dulcis* improving pancreatic function against islets cell apoptotic by Bax/Bcl-2/Caspase-3 pathway

Lin Dong^{a,1}, Mimi Chen^{c,1}, Zibao Huang^a, Yinfeng Tan^a, Caiyun Zhang^{a,b}, Shouwen Zhang^c, Yong Zhang^{a,**}, Xiaopo Zhang^{a,b,*}

^a Key Laboratory of Tropical Translational Medicine of Ministry of Education, Hainan Key Laboratory for Research and Development of Tropical Herbs, School of Pharmacy, Hainan Medical University, Haikou, 571199, China

^b Research Center for Drug Safety Evaluation of Hainan Province, Hainan Medical University, Haikou, 571101, China

^c Research Center for Traditional Chinese Medicine Resources and Ethnic Minority Medicine, Jiangxi University of Chinese Medicine, Nanchang, 330004, China

ARTICLE INFO

Handling Editor: Dr. Thomas Efferth

Keywords:

Scoparicol E

Antiapoptotic

Islet

Bax/Bcl-2/Caspase-3 pathway

Scoparia dulcis

ABSTRACT

Ethnopharmacological relevance: *Scoparia dulcis* has been identified as a significant ethnopharmacological substance in the Li, Zhuang, and Dai ethnic groups of China. Traditional medicine use *S. dulcis* to treat numerous illnesses, most notably diabetes. The considerable antidiabetic properties of this herbal remedy have been established by several clinical investigations and animal experiments. The islet is the intended target of *S. dulcis*, although the cause of its activity and mechanism for diabetes treatment is unclear. The diterpenoids from *S. dulcis* have been shown in the literature to have significant hypoglycemic efficacy and to protect islet cells *in vitro*. Diterpenoids may be the components of this herbal remedy that preserve islets, but further research is needed.

Aim of the study: This study was projected to investigate the new diterpenoid scoparicol E from *S. dulcis* and examined its islet-protective effect and the potential mechanism both *in vitro* and *in vivo*.

Methods: The structure of the novel diterpenoid scoparicol E was clarified by employing a wide range of spectroscopic methods. Using CCK-8 tests, cytotoxicity and antiapoptotic activity of scoparicol E were detected. Serum biochemical analysis and pathologic examination were performed to study the protective effect of scoparicol E against islet damage. The specific mechanism of action of scoparicol E was investigated through the mitochondrial membrane potential, Annexin V-FITC flow cytometry, and western blotting.

Results: Scoparicol E reduced MLD-STZ-induced hyperglycemia in mice and increased insulin and islet apoptosis. Scoparicol E effectively suppressed the Bax/Bcl-2/Caspase-3 pathway, according to the *in vivo* western blot investigation. Scoparicol E showed significant antiapoptotic action *in vitro*. We also showed that scoparicol E might prevent islet cells from dying by inhibiting the Bax/Bcl-2/Caspase-3 pathway. The Annexin V-FITC flow cytometry results revealed that MIN6 cell apoptosis was considerably decreased following scoparicol E intervention, showing anti-islet cell apoptosis action. Furthermore, the Caspase-3-mediated apoptosis pathway depends on cytochrome c and the potential of the mitochondrial membrane. Scoparicol E prevented the release of cytochrome c, restored the mitochondrial membrane potential, and prevented MIN6 cell apoptosis.

Conclusion: We demonstrated the new diterpenoid scoparicol E could protect islet cells apoptosis by modulating the Bax/Bcl-2/Caspase-3 pathway.

1. Introduction

Type 2 diabetes (T2DM) is a severe, long-term metabolic condition.

Insulin resistance and inadequate insulin secretion are the primary causes of its development. According to research, pancreatic islet cell apoptosis (PiβCA), which impairs islet cell function, contributes significantly to the development of T2DM. (You et al., 2022). Therefore, using

* Corresponding author. Key Laboratory of Tropical Translational Medicine of Ministry of Education, Hainan Key Laboratory for Research and Development of Tropical Herbs, School of Pharmacy, Hainan Medical University, Haikou, 571199, China.

** Corresponding author.

E-mail addresses: 935150242@qq.com (Y. Zhang), z_xp1412@163.com (X. Zhang).

¹ These authors contributed equally to this work.

Abbreviations

MLD-STZ	multiple low-dose streptozotocin
PiβCA	pancreatic islet cell apoptosis
TG	triglyceride
TC	total cholesterol
LDL-c	low density lipoprotein-cholesterol
HDL-c	high density lipoprotein-cholesterol
ALT	alanine aminotransferase
AST	aspartate aminotransferase
CCK-8	Cell Counting Kit-8
DMSO	dimethyl sulfoxide
FBS	fetal bovine serum

Caspase-3	cysteine aspartic acid specific protease-3
Bax	bcl2-associated x
Bcl-2	b-cell lymphoma-2
Cytc	cytochrome c
HR-ESI-MS	high-resolution electrospray ionization mass spectrometry
NMR	nuclear magnetic resonance
ECD	electronic circular dichroism
HSQC	¹ H-detected heteronuclear single quantum coherence
HMBC	¹ H-detected heteronuclear multiple bond correlation
¹ H- ¹ H COSY	¹ H- ¹ H correlation spectroscopy
NOESY	nuclear overhauser effect spectroscopy
IR	infrared spectroscopy

PiβCA inhibitors is a successful tactic to address bottleneck in diabetes treatment and blood sugar regulation.

For the purpose of conducting animal model studies on diabetes, streptozotocin (STZ) is frequently utilized (You et al., 2022; Fu et al., 2018). Oxidative stress increases after STZ treatment thus leading to β-cells damage and apoptosis. (Zhao et al., 2016; Al-Nahdi et al., 2018).

Caspase-3 controls an apoptotic signaling pathway that is triggered by circumstances, such as autophagy, mitochondrial dysfunction, oxidative stress, and ER stress, and it plays a role in the activation of apoptosis in β-cells. The cascade response to the activation of apoptotic receptors causes altered mitochondrial permeability, which in turn encourages the release of cytochrome c (Cytc) or apoptosis-inducing factors (AIFs), activates Caspase-3, and ultimately results in β-cells apoptosis (Wnęk et al., 2017). Proapoptotic proteins such as Bax and antiapoptotic proteins such as Bcl-2 maintain the body's homeostatic energy. When cells are stressed, a process associated with the triggering of caspase cascade reactions that regulate the mitochondrial pathway triggers apoptosis by increasing the expression of the protein Bax and decreasing the expression of the protein Bcl-2 (Ruiz-Vela et al., 2005).

Screening and identifying novel plant-derived active metabolites or lead compounds to protect islet cells from apoptosis has become a prevalent subject in the research field. Traditional uses of the ethnomedicinal *Scoparia dulcis* include the management of diabetes and its prevention (Dai et al., 2008; Wan et al., 2015; Pamunuwa et al., 2016). According to pharmacological and clinical studies, *S. dulcis* extracts have antidiabetic action by enhancing islet function (Sharma et al., 2015; Latha et al., 2004a,b). From these plants, diterpenoids, alkaloids, and flavonoids have been identified (Sun et al., 2021; Pamunuwa et al., 2016; Jiang et al., 2021). At a dose of 10 mg/kg, the diterpenoid scopiric acid D (SAD) significantly reduced blood glucose levels in diabetic rats. (Latha et al., 2009). Additionally, SAD somewhat shields pancreatic islet cells against damage and can enhance their production *in vitro* (Latha et al., 2009). Three novel labdane diterpenoids were identified and structurally described in our earlier research. Notably, three novel diterpenoids, scoparic acids A, B, and E, were able to reduce palmitate's effect on MIN6 cell induction of viability (Sun et al., 2020; Zhang et al., 2020). As a result, it is possible that the diterpenoids discovered in *S. dulcis* shield islet cells. This study extracted scoparicol E from *S. dulcis* and a novel labdane diterpenoid with anti-islet cell apoptotic action. Nuclear magnetic resonance (NMR) spectroscopy in 1D and 2D, high-resolution electrospray ionization-mass spectrometry (HR-ESI-MS), and electronic circular dichroism (ECD) were used to determine the structure of scoparicol E from *S. dulcis*. Additionally, *in vitro* and *in vivo* studies on scoparicol E's antiapoptotic properties and mechanism were conducted.

2. Materials and methods

2.1. Plant material

The plant name was checked in <http://www.worldfloraonline.org>. *Scoparia dulcis* is the accepted name of a species in the genus *Scoparia* (family Plantaginaceae). In September 2020, the whole plant of *S. dulcis* was collected from Danzhou City, Hainan Province, China. A voucher specimen of this plant (No. SD202009) has been labelled and deposited at the herbarium of School of Pharmacy, Hainan Medical University.

2.2. Isolation of scoparicol E

The new diterpenoid scoparicol E was isolated from the petroleum ether fraction of 95% ethanol extract of *S. dulcis* by silica gel chromatography and semi-preparative high performance liquid chromatography, and its structure was identified based on spectroscopic techniques. The specific separation procedure is depicted in detail in the supplementary material.

2.3. Cell culture

At 37 °C with 5% CO₂, MIN6 cells were grown in RPMI-1640 media with 10% FBS, 50 M β-mercaptoethanol, and 1% penicillin-streptomycin. In 96-, 12-, or 6-well plates, logarithmic-growth MIN6 cells were plated and grown for 24 h prior to the addition of drugs. MIN6 cells were purchased from Anhui Differential Gene Technology Co., Ltd.

2.4. CCK-8 assay

MIN6 cells were collected, counted on a hemocytometer plate, and then injected at 1×10^4 in a 96-well plate using a logarithmically developed culture. The cells were incubated for 24 h at 37 °C with 5% CO₂ in five replicate wells that had a final volume of 100 μL each. Drug-treated cells for 24 h. The drug-containing supernatant was aspirated in accordance with (Xia et al., 2020), and 110 μL of the CCK-8 working media were supplied to each well. One hundred microliters of RPMI-1640 medium and 10 μL of CCK-8 were incubated at 37 °C for 2 h. The following equation was used to calculate the cell viability using the absorbance value at 450 nm (OD₄₅₀): $[(OD_{450} \text{ for drug administration group} - OD_{450} \text{ for blank group}) / (OD_{450} \text{ for control group} - OD_{450} \text{ for blank group})] \times 100\%$.

2.5. Design of animal experiments

Male C57BL/6J mice aged four weeks were purchased from Gem-Pharmatech Co., Ltd. in Jiang Su, China. The mice were kept at the Animal Research Center of Hainan Medical College with unrestricted access to water and food (common chow) at room temperature (25 ±

0.5 °C) and a humidity of approximately 70% on a 12-h light-dark cycle. The Animal Research Center at the Hainan Medical College served standard fare. The Hainan Medical College Animal Ethics Committee's guidelines were followed in conducting the animal experimentation operations. The Hainan Medical College Medical Ethics Committee granted approval for the experimental protocol (syxk-2017-0013). According to earlier research, animals were handled and used in studies (Shehata et al., 2011). The mice were randomly assigned to one control group, one STZ treatment group, and two STZ + scoparicol E intervention groups, each consisting of 10 mice, after 7 days of acclimation to baseline chow. STZ injections (50 mg/kg) into the peritoneum were administered to the mice (dissolved in a pH 4.5 sodium citrate buffer) over the course of five days, whereas the control group was given injections of the same amount of citric acid-sodium citrate buffer at a rate of 10 mL/kg. Additionally, mice in the scoparicol E intervention groups received scoparicol E for 7 days either at low doses of 5 mg/kg or high doses of 10 mg/kg. The control and STZ groups were given vegetable oil, the vehicle for scoparicol E. The animals were fed a basic diet and observed for 14 d. Serum samples were extracted from the collected blood for further analysis. Following the collection of blood, tissue samples were gathered for further examination. Serum samples were extracted from the collected blood for further analysis. Tissue samples were obtained after blood collection for further examination.

2.5.1. Intraperitoneal glucose tolerance test (IPGTT)

According to a prior procedure (Zhang et al., 2022), all test animals were fasted for 6 h. The value for time point 0 was taken from the measured fasting blood glucose at this time. After measuring the blood glucose levels 30, 60, 120, and 180 min after glucose ingestion, curves were created using the data.

2.5.2. Biochemical analysis

Whole blood was centrifuged for 15 min at 4 °C and 3500 rpm after being stored at room temperature for 4 h. The supernatant was then poured into a fresh centrifuge tube. The kit's instructions were followed while measuring the TG, TC, LDL-c, HDL-c, ALT, and AST levels in mouse serum. The concentrations of serum insulin were measured using an insulin enzyme-linked immunosorbent assay kit.

2.5.3. Histopathological analysis of the pancreas

Mice were swiftly dissected, and the pancreas was promptly removed, rinsed with saline, and sliced into an appropriate size before being preserved in 10% paraformaldehyde tissue fixative for 48 h. After being completely fixed, slices of the tissues that were regularly paraffin-embedded (5 µm) were then coated with hematoxylin-eosin (H&E) and photographed under a light microscope.

2.5.4. Mitochondrial isolation and western blotting

Following the guidelines provided by the manufacturer, mitochondria were isolated from MIN6 cells using the Mitochondrial Isolation Kit. MIN6 cells were centrifuged at 800×g for 10 min at 4 °C after they had been homogenized in 1 mm PMSF in cryogenic mitochondrial isolation buffer. Afterward, the supernatant was put into a new centrifuge tube and spun for 15 min at 11,000×g at 4 °C. The supernatant contained cytoplasmic proteins without mitochondria, while the deposit contained mitochondria (Zhang et al., 2018).

Total protein was extracted from pancreatic tissue or cells. Cellular proteins were extracted using RIPA lysis buffer (1 × protease inhibitor and 1 × phosphoprotease inhibitor), quantified, separated by polyacrylamide gel electrophoresis, transferred to membranes, incubated overnight with primary antibodies (1:1000 dilution in 5% skim milk powder/1 × TBST), washed with 1 × TBST, incubated for 1 h with fluorescent secondary antibodies, washed with 1 × TBST and then stained using the ECL kit. The proteins Caspase-3, Bax, Bcl-2 and Cyt c were detected, and using ImageJ software, the grayscale values of the protein bands were examined.

2.6. Flow cytometry

After therapy, the cells were separated from one another using trypsin containing 0.25% EDTA, and digestion was stopped by adding an equivalent amount of fresh RPMI-1640 medium. After being washed twice with PBS and resuspended in 200 µL of binding buffer, the cells were retrieved by centrifuging them at 1000×g for 5 min at room temperature. The samples were then gently mixed with 5 µL of 7-AAD dye before being kept at ambient temperature and protected from light for 15 min. Next, 400 µL of binding buffer was utilized following staining. The samples were then passed through a 200 mesh filter.

2.7. Mitochondrial membrane potential

As previously mentioned, MIN6 cells were inoculated in 6-well plates, and they were then exposed to 300 µM PA as well as different doses of scoparicol E for 24 h, according to (Cao et al., 2021). After 24 h, the supernatant containing the medication was removed and discarded. Next, PBS was used to wash the cells. Then, 1 mL each of RPMI-1640 medium and JC-1 staining buffer were supplied. The supernatant was removed after 20 min of incubation at 37 °C, and JC-1 staining buffer (1 ×) was used to wash the cells thereafter. The mitochondrial membrane potential of the cells was then assessed by fluorescence microscopy after the addition of 2 mL of RPMI-1640 medium to the cells.

2.8. Statistical analysis

The mean and standard deviation (S.D.) of the *in vitro* data are displayed in at least three separate studies. The findings of the *in vivo* tests are shown as the mean ± S.D. of 10 mice per group. GraphPad Prism 8.0 software was used for processing the information and analysis of statistics. To find differences between groups, a one-way ANOVA was employed, and $p < 0.05$ was deemed statistically significant.

3. Results

3.1. Structural analysis of scoparicol E

White powder ($[\alpha]_D^{20}$ 57 (c 0.1, CH₃OH)). Its molecular formula

was established as C₂₇H₃₆O₅ from its HR-ESI-MS data (m/z 463.2456 ([M+Na]⁺, calcd. For 463.2455), and the calculated unsaturation is 10. The IR absorption bands at 3463, 1712 and 1452 cm⁻¹ indicated the presence of hydroxyl, carbonyl, and phenyl groups in the molecule. The

Table 1
NMR data of scoparicol E (600 MHz for ¹H and 150 MHz for ¹³C) in CDCl₃.

No.	δ _H (multi, J in Hz)	δ _C	No.	δ _H (multi, J in Hz)	δ _C
1α	1.54 (m)	38.2	13	–	146.2
1β	1.15 (m)		14	–	200.0
2α	1.74 (m)	18.7	15	4.57 (brs)	64.9
2β	1.55 (m)		16a	5.86 (brs)	125.3
3α	1.19 (m)	38.5	16a	5.94 (brs)	
3β	1.58 (m)		17	4.76 (brs)	113.5
4	–	38.7	18a	3.14 (d, 10.9)	71.4
5	2.03 (d, 2.4)	41.5	18b	3.60 (d, 10.9)	
6	5.63 (q, 2.8)	71.7	19	0.90 (s)	20.5
7α	2.43 (m)	37.1	20	1.46 (s)	26.1
7β	2.37 (m)		1'	–	131.0
8	–	144.3	2', 6'	8.04 (dd, 8.2, 1.2)	129.8
9	1.85 (m)	57.6	3', 5'	7.44 (dd, 8.2, 7.4)	128.6
10	–	38.9	4'	7.55 (td, 7.4, 1.2)	133.1
11a	1.46 (m)	25.3	7'	–	166.5
11b	1.76 (m)				
12a	2.07 (m)	30.1			
12b	2.21 (m)				

^1H NMR spectral data (Table 1) displayed signals for a benzoyl group [δ_{H} 7.44 (2H, dd, $J = 8.2, 7.4$ Hz, H-3', 5'), 7.55 (1H, td, $J = 7.4, 1.2$ Hz, H-4'), 8.04 (2H, dd, $J = 8.2, 1.2$ Hz, H-2', 6')]; two terminal double bonds [δ_{H} 5.86 (1H, brs), 5.94 (1H, brs), and 4.76 (2H, brs)]; four oxygenated methylenes at δ_{H} 3.14 (1H, d, $J = 10.9$ Hz, H-18), 3.60 (1H, d, $J = 10.9$ Hz, H-18), and δ_{H} 4.57 (2H, brs, H-15); and one oxygenated methine at δ_{H} 5.63 (1H, q, $J = 2.8$ Hz, H-6); two methyl groups at δ_{H} 0.90 (3H, s) and 1.46 (3H, s). The ^{13}C NMR data (Table 1), in conjunction with the HSQC experimental data, resolved 27 carbon resonances attributable to one benzoyl group (δ_{C} 166.5, 133.1, 129.8, 129.8, 128.6, 128.6, and 131.0), one ketocarbonyl group [δ_{C} 200.0 (C-14)], four olefinic carbons (δ_{C} 144.3, 146.2, 113.5, and 125.3), two oxygenated methylenes (δ_{C} 64.9 and 71.4), and one oxygenated methine (δ_{C} 71.7), along with six methylenes (δ_{C} 18.7, 25.3, 30.1, 37.1, 38.2, and 38.5), two methines (δ_{C} 41.5 and 57.6), two quaternary carbons (δ_{C} 38.7 and 38.9), and two methyls (δ_{C} 20.5 and 26.1).

By these analyses, the compound possessed a labdane diterpenoid skeleton. The ^1H - ^1H COSY correlations helped to build three structural units of H-1/H-3, H-5/H-7 and H-9/H-11/H-12, as shown in Fig. 2. The connections of these different units were determined by analyzing the HMBC spectrum (Fig. 2). Specifically, the HMBC cross-peaks of H₃-20 to C-1, C-5, C-9 and C-10; H₃-19 to C-3, C-4, C-5 and C-18; H₂-17 to C-7, C-8 and C-9; H-9 to C-8, C-10, C-11, and C-12; H₂-16 to C-12, C-13 and C-14; and H₂-15 to C-14, C-13 and C-16 confirmed that the carbonyl group was located at position C-14 of scoparicol E. Based on the HMBC correlation from H-6 to C-7' together with the chemical shifts for H-6 (δ_{H} 5.63) and C-7' (δ_{C} 166.4), where the benzoyl group is connected to C-6. The above data indicate that the structure of scoparicol E is similar to that of scoparicol C (Li et al., 2022), with the difference being that the connection at C-14 of scoparicol E is a carbonyl group. In summary, the planar structure of scoparicol E was determined.

The relative stereochemistry of scoparicol E was determined based on analysis of NOESY data (Fig. 2). The NOESY correlations between H₃-19 and H₃-20 and H-2'(6') indicated that Me-19 and Me-20 were β -axially and H-6 was α oriented. Then, the NOESY correlation of H-5 and H-6 indicated that H-5 was α -axially bonded. Furthermore, the side chain at C-9 was determined to be in the α -axial direction based on the NOESY correlation between H₃-20 and H-9, as shown in Fig. 2. The absolute configuration of scoparicol E was established to be (4*R*,5*R*,6*R*,9*R*,10*R*) by comparing its experimental ECD spectrum with the calculated spectrum (Dong et al., 2023; Sybyl Software, 2013; Neese, F., 2012; Stephens et al., 2010.) (Fig. 3). Thus, the structure of scoparicol E was established as illustrated in Fig. 1. After systematic document retrieval, scoparicol E was found to be a new compound that has not yet been reported.

3.2. Antiapoptotic activity of scoparicol E in MIN6 cells

Before assaying its antiapoptotic activity, the toxicity of scoparicol E to MIN6 cells was determined using the CCK-8 assay. We first added different concentrations of scoparicol E to MIN6 cells in the intervention treatment group and assayed the cellular activity after 24 h. The CCK-8 results showed that scoparicol E concentrations of 12.5 μM and below were not toxic to MIN6 cells, whereas concentrations above 25 μM resulted in a decrease in MIN6 cell viability (Fig. 4A). Therefore, we chose concentrations of 12.5 μM and below to detect the antiapoptotic activity of scoparicol E in MIN6 cells. Incubation with 300 μM PA for 24 h induced apoptosis in normal MIN6 cells (98.21 ± 4.237 vs. $58.68 \pm 4.879\%$, $p < 0.01$). Compared with the model group (PA), the cell apoptosis rates following treatment with various doses of scoparicol E (0.39, 0.78, 1.56, 3.125, 6.25, and 12.5 μM) were significantly higher after 24 h of treatment (76.13 ± 7.119 , 84.12 ± 1.216 , 93.69 ± 4.610 , 100.2 ± 2.348 , 107.0 ± 4.47 and $107.1 \pm 2.40\%$, $p < 0.01$), with a dose-dependent enhancement in cell viability. These results suggested that scoparicol E could effectively ameliorate PA-induced apoptosis in MIN6

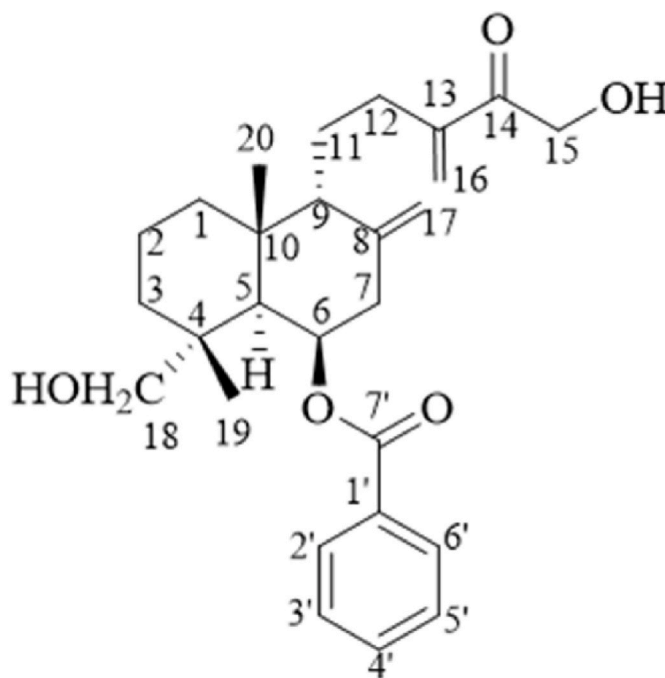


Fig. 1. Structure of scoparicol E.

cells (Fig. 4B).

3.3. Scoparicol E reduces STZ-induced hyperglycemia and improves islet damage in mice

In mice, the scoparicol E intervention significantly decreased glucose tolerance, fasting blood glucose levels, and final blood glucose levels (Fig. 5A–C). The final blood glucose level in the mice in the scoparicol E (5 mg/kg) group was 14.23 ± 5.74 mM, which was lower than the STZ group's final blood glucose level. The STZ group of mice had considerably larger pancreatic weights (0.16 ± 0.026 g) and pancreatic weight/body weight ratios (0.0076 ± 0.0014) than the normal group (Fig. 5D and E). The pancreatic weight and pancreatic weight/body proportions of the mice were dramatically decreased to 0.098 ± 0.011 g and 0.0055 ± 0.0008 , respectively, following scoparicol E (5 mg/kg) intervention (Fig. 5D and E). The animals in the normal group exhibited a clear histological structure with spherical islets and clear boundaries with the surrounding glands, according to histopathological examination of the pancreas H&E staining pictures. The islets in the STZ group showed uneven shapes and fuzzy boundaries, fewer cells, and noticeably more vacuoles than the islets in the control group. These findings suggested that scoparicol E therapy might lessen the islet damage caused by STZ in mice (Fig. 5F). In comparison to the normal group (3.16 ± 0.054 IU/L), the insulin level in the STZ group (3.04 ± 0.10 IU/L) was substantially lower. The insulin level increased after scoparicol E (5 mg/kg) therapy to 3.18 ± 0.04 IU/L (Fig. 5L).

Additionally, the mice in the STZ group had serum TC and LDL-C concentrations of 3.61 ± 0.20 and 1.51 ± 0.11 mM, respectively, and serum HDL-C concentrations of 3.95 ± 0.42 mM. In STZ mice, there were higher serum lipid levels overall compared to the control group ($p < 0.05$) (Fig. 5G–I). The TC and LDL-C levels were decreased to 2.80 ± 0.43 and 1.31 ± 0.15 mM, respectively, after scoparicol E (5 mg/kg) therapy (Fig. 5G and H). These results indicated that scoparicol E could improve diabetes-associated lipid metabolism. The STZ mice had serum ALT and AST concentrations of 47.88 ± 4.37 and 45.96 ± 9.08 U/L, respectively, which were much greater than those in the control mice. (ALT, 34.95 ± 2.70 and AST, 42.97 ± 2.96 U/L) (Fig. 5J and K). Scoparicol E (5 mg/kg) therapy decreased serum ALT and AST levels to

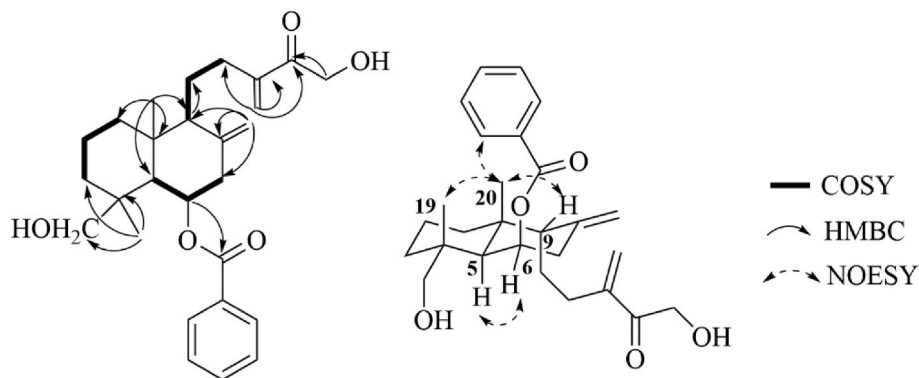


Fig. 2. Key HMBC, ^1H - ^1H COSY and NOESY correlations of scoparicol E.

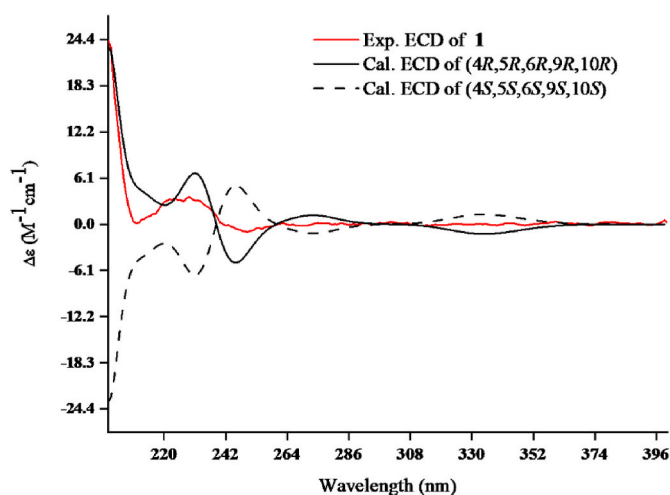


Fig. 3. Calculated ECD and experimental ECD spectra (200–400 nm) of scoparicol E.

36.51 ± 6.36 and 25.98 ± 6.98 U/L, respectively, which were substantially less than those of the mice in the STZ group. These findings suggest that scoparicol E reduces STZ-induced hyperglycemia and improves islet damage in mice.

3.4. Scoparicol E inhibits STZ-induced apoptosis in mouse pancreatic tissue

We carried out western blot experiments on mouse pancreatic tissues

to more thoroughly investigate the probable mechanism of the anti-apoptotic action of scoparicol E in mice with diabetes. The results showed that in STZ mouse pancreatic tissues, Bax and cleaved Caspase-3 protein expression was considerably greater and Bcl-2 protein expression was significantly lower than that in the control group (Fig. 6A).

When compared to the STZ group, the Bax/Bcl-2 ratio was 1.55 ± 0.018 , and the expression of cleaved Caspase-3 protein was 1.85 ± 0.42 (Fig. 6B and C). After scoparicol E administration, the Bax/Bcl-2 ratio in mouse pancreatic tissue was reduced to 1.22 ± 0.073 and 1.27 ± 0.052 (Fig. 6B and C), respectively, and the protein expression of cleaved Caspase-3 decreased to 1.11 ± 0.27 and 1.15 ± 0.13 , respectively. According to all of these findings, scoparicol E prevents STZ from inducing apoptosis in mouse pancreatic tissue.

3.5. Scoparicol E inhibits the Bax/Bcl-2/caspase-3 pathway to inhibit islet cell apoptosis

To further investigate the molecular basis of scoparicol E's anti-apoptotic actions *in vitro*, we used western blotting to explore the components of the traditional Bax/Bcl-2/Caspase-3 pathway. Bax and cleaved Caspase-3 protein expressions were considerably greater in the PA group than in the CK group, but Bcl-2 protein expression was substantially lower (Fig. 7A). The Bax/Bcl-2 ratio and the protein expression of cleaved Caspase-3 were 12.78 ± 3.29 and 1.57 ± 0.13 , respectively (Fig. 7B and C). After treatment with scoparicol E at various doses (3.125, 6.25, and 12.5 M), the expression of cleaved Caspase-3 protein decreased to 6.11 ± 0.73 , 9.34 ± 1.30 , and 3.92 ± 0.27 , respectively, and the ratio of Bax/Bcl-2 decreased to 1.15 ± 0.10 , 1.20 ± 0.17 , and 0.75 ± 0.23 , respectively (Fig. 7B and C).

To gauge the release of Cytc, we performed a western blot because it has been shown that this protein is released from the mitochondria in

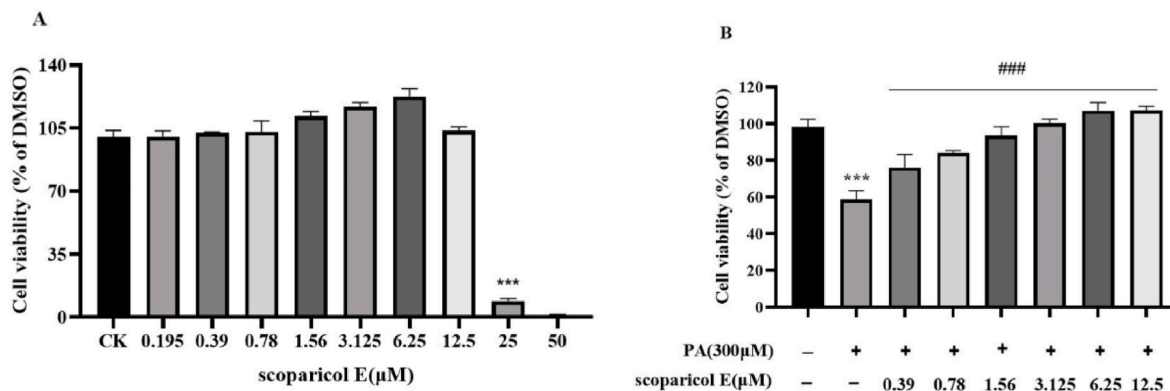


Fig. 4. Antiapoptotic activity of scoparicol E in MIN6 cells. (A) and (B) CCK-8 assay data. All the statistics show the means $\pm$ SDs. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. DMSO; # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$ vs. PA.

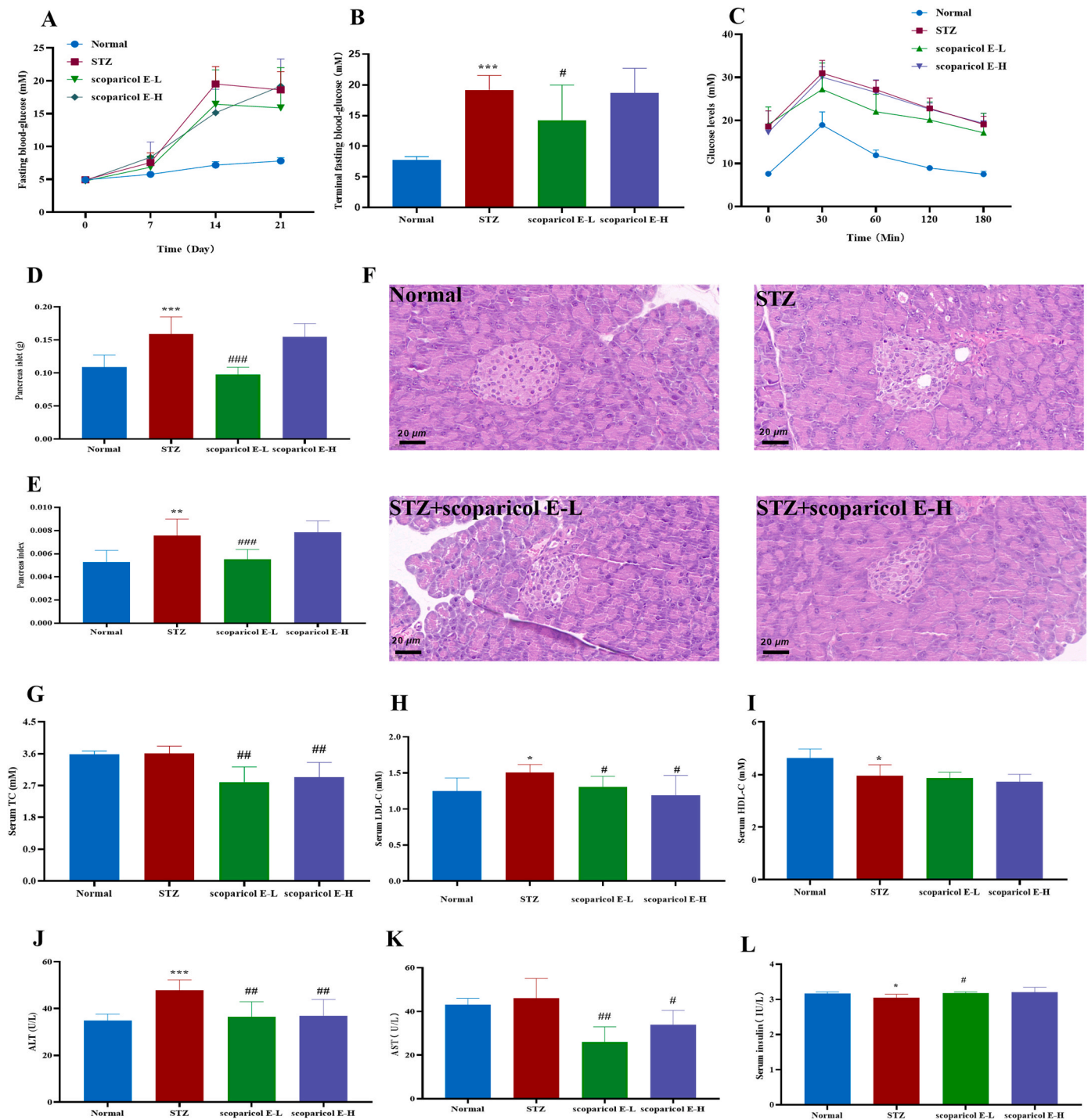


Fig. 5. Scoparicol E decreased the STZ-induced hyperglycemic status and ameliorated islet injury in mice. (A) Fasting blood glucose. (B) Terminal fasting blood glucose. (C) IPGTT. (D) Pancreatic weight. (E) Pancreatic index. (F) Images of the pancreas that represent each animal type, as seen in one H&E-stained photograph each. (G) Total cholesterol (TC). (H) Low-density lipoprotein cholesterol (LDL-C). (I) High-density lipoprotein cholesterol (HDL-C). (J) Serum ALT. (K) Serum AST. (L) Serum insulin levels. Each data point represents the mean $\pm$ SD ($n = 10$ per group). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. normal; # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$ vs. model (STZ mice).

conjunction with Caspase-3 activation and the beginning of apoptosis (Zhang et al., 2018). In contrast to CK, PA-induced apoptosis in MIN6 cells resulted in a 0.43 ± 0.020 decrease in the protein expression of Cytc in mitochondria and a 1.95 ± 0.032 rise in the protein expression of Cytc in the cytoplasm. Notably, scoparicol E (3.125, 6.25, and 12.5 M) reversed Cytc protein production that was stimulated by PA in the mitochondria and cytoplasm.

With the use of flow cytometry and Annexin V-FITC single labeling,

PA-induced apoptosis in MIN6 cells was assessed. When opposed to the control (8.44%), the proportion of MIN6 cells that underwent apoptosis as a result of PA (22.86%) was substantially higher. Additionally, the level of apoptosis in MIN6 cells in the scoparicol E group was much lower (13.26%) (Fig. 7G). Thus, scoparicol E might lessen the apoptosis that PA induces in MIN6 cells.

A considerable reduction in JC-1 aggregates (red fluorescence) and a significant rise in JC-1 monomers (green fluorescence) were observed in

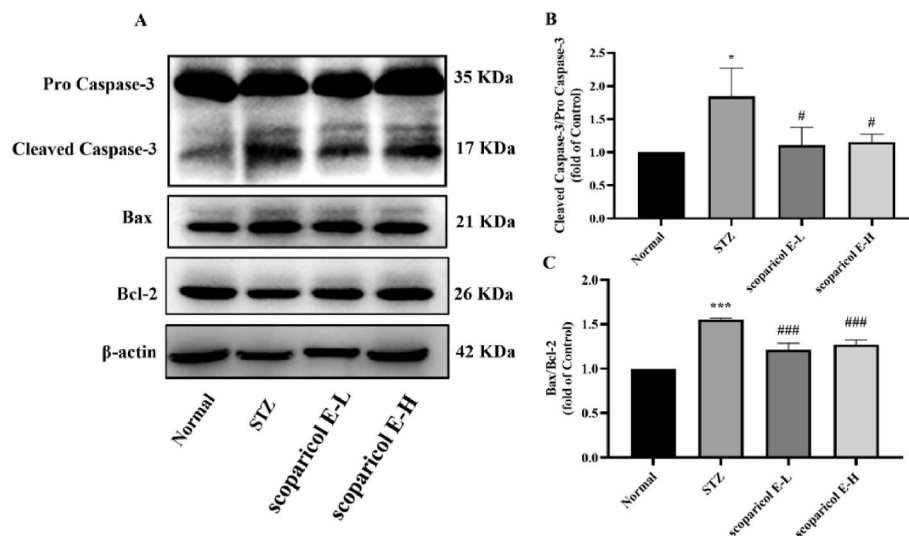


Fig. 6. Scoparicol E inhibits STZ-induced apoptosis in mouse pancreatic tissue. Each data point represents the mean $\pm$ SD ($n = 10$ per group). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. normal; # $p < 0.05$, ### $p < 0.01$, #### $p < 0.001$ vs. model (STZ mice).

MIN6 cells after PA treatment (Fig. 7H). The red fluorescence was effectively increased, while the green fluorescence was diminished when scoparicol E and PA were administered together for 24 h. Therefore, scoparicol E restored the potential of the mitochondrial membrane. These data indicate that scoparicol E suppresses the Bax/Bcl-2/Caspase-3 pathway and offers protection from this islet cell apoptosis molecular mechanism.

4. Discussion

In this study, scoparicol E was discovered to have strong anti-PI3CA activity both *in vitro* and *in vivo*, revealing its anti-T2DM potential and providing new clues for the study of the anti-T2DM components of *S. dulcis*.

T2DM, one of the most chronic metabolic illnesses with the quickest growth rate, poses a severe threat to people's lives and health. The primary pathophysiology is pancreatic β -cells malfunction, which results in insulin resistance and inadequate insulin production (You et al., 2022). Consequently, T2DM is represented by a steady reduction in pancreatic islet function and a gradual decline in pancreatic islet function (Drucker, 2018). Metformin, glibenclamide, pioglitazone and repaglinide are the current commercially available hypoglycemic agents, and although they can reduce blood glucose levels in patients with T2DM, they are still unable to satisfy clinical needs, especially in patients with diabetes accompanied by hyperglycemia or hyperlipidemia stimulation induced by islet apoptosis. Therefore, the key to effectively treating T2DM is utilizing novel anti-PI3CA drugs to play an essential part in pancreatic islet β -cells protection, slowing the decline in islet β -cells numbers, and alleviating pancreatic islet cell hypoplasia, in addition to lowering blood glucose and managing other complications (Mathiesen et al., 2019). Ethnic drugs in the prevention and treatment of diabetes have the unique advantage of gentle actions and multiple targets, especially because ethnic drugs contain structurally unique and active antidiabetic ingredients, which has made them an active area of modern medical research (Sharma et al., 2015).

The ethnomedicine *S. dulcis* belongs to the genus *Scoparia* and family Plantaginaceae, and is a complete plant that is historically used to prevent and cure diabetes (Dai et al., 2008; Wan et al., 2015; Pamunuwa et al., 2016). The *S. dulcis* diterpenoid is a unique component, and its structure and pharmacological activity, especially its hypoglycemic effect, have received special attention (Sun et al., 2021; Pamunuwa et al., 2016; Jiang et al., 2021). It was reported in the literature that the diterpenoids from *S. dulcis* had clear hypoglycemic activity and a

protective effect on islet cells *in vitro* (Latha et al., 2009), which indicated that diterpenoids may be the islet protective components of this herbal medicine. Previous studies on the antidiabetic effect of *S. dulcis* have measured only the efficacy of extracts and the *in vitro* hypoglycemic activity of some monomer components (Pamunuwa et al., 2016; Jiang et al., 2021). However, its active antidiabetic components and their islet protection mechanism are still unclear.

In this study, we obtained a novel diterpene, scoparicol E, from *S. dulcis* using modern separation techniques to determine its structure. Our *in vitro* experimental findings demonstrated that scoparicol E significantly inhibited MIN6 cell apoptosis. The *in vivo* results showed that scoparicol E significantly reversed the elevations in pancreatic weight, pancreatic weight index, fasting glucose, terminal glucose, insulin levels, glucose tolerance, and islet damage; thus, improved the hyperglycemic state and islet function in STZ mice compared to STZ model mice.

We also demonstrated how scoparicol E prevents PA from inducing apoptosis in MIN6 cells and STZ from inducing pancreatic islet damage. Bax was one of the first proteins discovered to trigger apoptosis (Zheng et al., 2016; Karch et al., 2015), and Bcl-2 is a key member of the familial group of antiapoptotic proteins (Sharma et al., 2017; Liu et al., 2019). Together, Bax and Bcl-2 mediate Caspase-3 activation and cleavage (Wang et al., 2012). Both PA and STZ decreased Bcl-2 expression and enhanced Bax expression, which resulted in an elevated Bax/Bcl-2 ratio, leading to significant activation of Caspase-3. However, scoparicol E treatment reversed the PA- and STZ-induced changes in the Bax/Bcl-2 ratio and protein levels of activated and cleaved Caspase-3. The production of mitochondrial apoptotic factors such as Cytc or apoptosis-inducing factor (AIF), which then activates Caspase-3 and causes apoptosis, is another way that the mitochondrial pathway contributes to the initiation and progression of apoptosis (Li et al., 1997). Cytc is released from the mitochondria into the cytoplasm after induction by PA and STZ, but the release of Cytc is reversed after scoparicol E treatment. Scoparicol E effectively prevented apoptosis and restored the mitochondrial membrane potential, according to results from flow cytometry and the mitochondrial membrane potential assay. The above findings suggest that scoparicol E can exert islet-protective effects by modulating the Bax/Bcl-2/Caspase-3 pathway.

5. Conclusions

In this paper, we have discovered a novel diterpenol scoparicol E with pancreatic island protective activity from *S. dulcis*. Scoparicol E

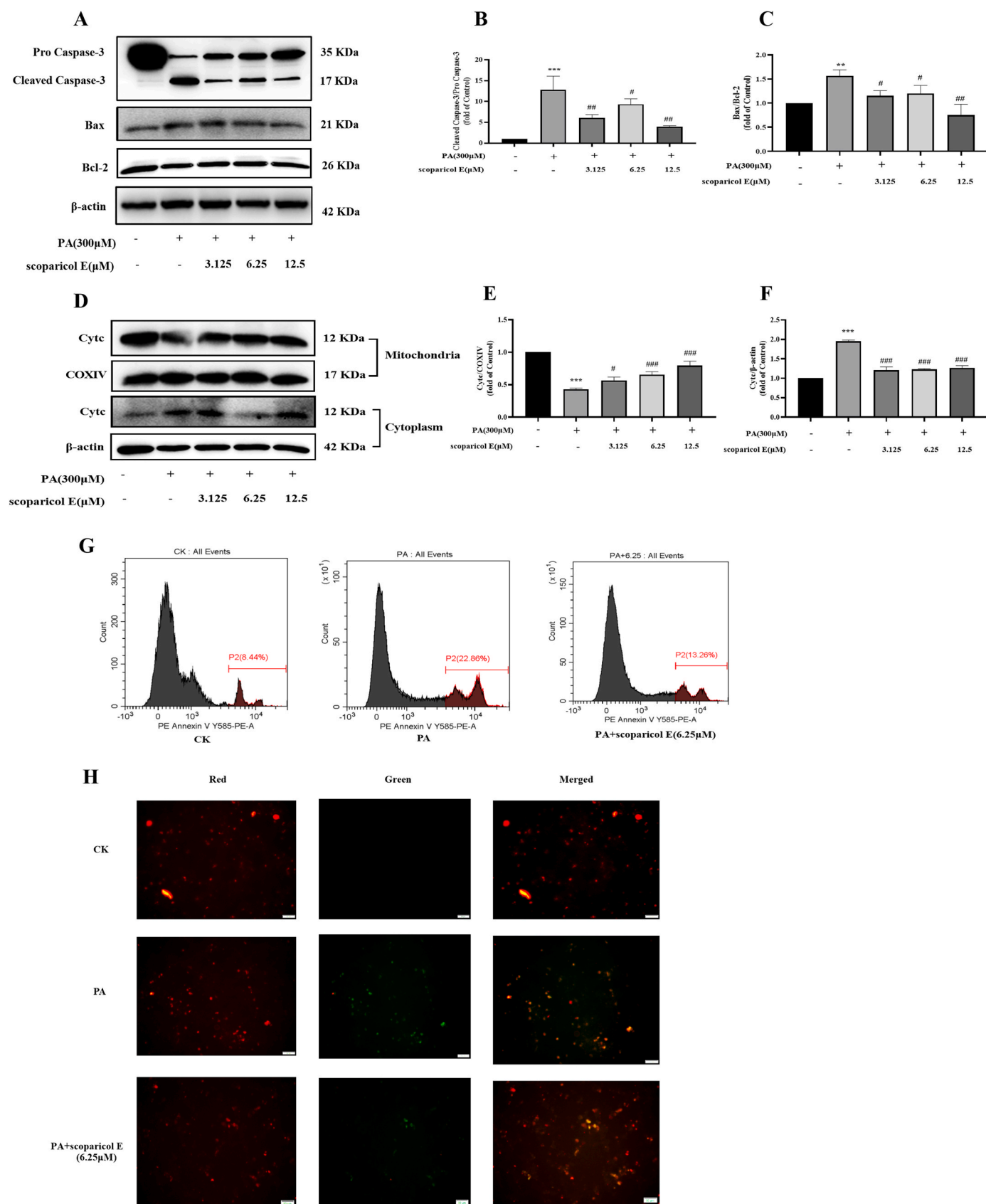


Fig. 7. Scoparicol E inhibits the Bax/Bcl-2/Caspase-3 pathway and protects against this molecular mechanism of pancreatic islet cell apoptosis. (A) and (D) Representative images of blots. (B) Fold changes in cleaved Caspase-3/pro Caspase-3. (C) Fold changes in Bax/Bcl-2. (E) Fold changes in Cytochrome c/COXIV. (F) Fold changes in Cytochrome c/β-actin. (G) Annexin V-FITC/PI single staining. (H) Mitochondrial membrane potential. All the statistics show means ± SDs * $p < 0.05$ or ** $p < 0.01$ vs. DMSO; # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$ vs. PA.

may inhibit the apoptosis of pancreatic islet cells via modulation of the Bax/Bcl-2/Caspase-3 pathway. This discovery serves as a foundation for future research on natural lead compounds with antidiabetic activity to prevent islet cell apoptosis and to treat diabetes.

Funding sources

The National Natural Science Foundation of China (82060778) provided financial assistance for this project.

CRediT authorship contribution statement

Lin Dong: Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft. **Mimi Chen:** Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft. **Zibao Huang:** Formal analysis, Investigation, Writing – review & editing. **Shouwen Zhang:** Writing – review & editing. **Yong Zhang:** Data curation, Formal analysis, Writing – review & editing. **Xiaopo Zhang:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.117571>.

References

- Al-Nahdi, A.M.T., John, A., Raza, H., 2018. Cytoprotective effects of N-acetylcysteine on streptozotocin-induced oxidative stress and apoptosis in RIN-5F pancreatic β -cells. *Cell. Physiol. Biochem.* 51, 201–216.
- Cao, Y., Chen, Z.W., Hu, J.J., Feng, J., Zhu, Z.J., Fan, Y.Q., Lin, Q.X., Ding, G.H., 2021. Mfn2 regulates high glucose-induced MAMs dysfunction and apoptosis in podocytes via PERK pathway. *Front. Cell Dev. Biol.* 9, 769213.
- Dai, H.F., Mei, W.L., 2008. *Li Ethnic Medicinal Records* (Volume I). China Science and Technology Press, pp. 218–219.
- Drucker, D.J., 2018. Mechanisms of Action and Therapeutic Application of Glucagon-like Peptide-1. *Cell metabolism* 27 (4), 740–756.
- Fu, J.Q., Zheng, H.Z., Cui, Q., Chen, C.J., Bao, S.M., Sun, J., Li, L., Yang, B., Wang, H.H., Hou, Y.Y., Xu, Y.Y., Xu, Y.H., Zhang, Q., Pi, J.B., 2018. Nfe2l1-silenced insulinoma cells acquire aggressiveness and chemoresistance. *Endocr. Relat. Cancer* 25 (3), 185–200.
- Jiang, Z.K., Sung, J.H., Wang, X.Y., Zhang, Y.Y., Wang, Y.M., Zhou, H.F., Wen, L., 2021. A review on the phytochemistry and pharmacology of the herb *Scoparia dulcis* L. for the potential treatment of metabolic syndrome. *RSC Adv.* 11, 31235–31259.
- Karch, J., Molkentin, J.D., 2015. Regulated necrotic cell death: the passive aggressive side of Bax and Bak. *Circ. Res.* 116 (11), 1800–1809.
- Latha, M., Pari, L., Sitasawad, S., Bhone, R., 2004a. Insulin-secretagogue activity and cytoprotective role of the traditional antidiabetic plant *Scoparia dulcis* (Sweet Broomweed). *Life Sci.* 75 (16), 2003–2014.
- Latha, M., Pari, L., Sitasawad, S., Bhone, R., 2004b. *Scoparia dulcis*, a traditional antidiabetic plant, protects against streptozotocin induced oxidative stress and apoptosis in vitro and in vivo. *J. Biochem. Mol. Toxicol.* 18 (5), 261–272.
- Latha, M., Pari, L., Ramkumar, K.M., Rajaguru, P., Suresh, T., Dhanabal, T., Sitasawad, S., Bhone, R., 2009. Antidiabetic effects of scoparic acid D isolated from *Scoparia dulcis* in rats with streptozotocin-induced diabetes. *Nat. Prod. Res.* 23 (16), 1528–1540.
- Li, P., Nijhawan, D., Budihardjo, I., Srinivasula, S.M., Ahmad, M., Alnemri, E.S., Wang, X., 1997. Cytochrome c and dATP-dependent formation of Apaf-1/caspase-9 complex initiates an apoptotic protease cascade. *Cell* 91 (4), 479–489.
- Li, Y.P., Wu, D.X., Ye, T., Zhang, H., 2022. Cytotoxic diterpenoids from the aerial parts of *Scoparia dulcis*. *Phytochem. Lett.* 49, 21–26.
- Liu, L., Liang, C., Mei, P.C., Zhu, H., Hou, M.L., Yu, C.L., Song, Z.B., Bao, Y.L., Huang, Y. X., Yi, J.W., Wang, S.Y., Wu, Y., Zheng, L.H., Sun, Y., Wang, G.N., Huo, M.X., Yang, S.N., Sun, L.G., Li, Y.X., 2019. Dracorhodin perchlorate protects pancreatic β -cells against glucotoxicity- or lipotoxicity-induced dysfunction and apoptosis *in vitro* and *in vivo*. *FEBS J.* 286 (18), 3718–3736.
- Mathiesen, D.S., Bagger, J.I., Bergmann, N.C., Lund, A., Christensen, M.B., Vilsbøll, T., 2019. The effects of dual GLP-1/GIP receptor agonism on glucagon secretion-A review. *Int. J. Mol. Sci.* 20 (17), 4092.
- Neese, F., 2012. The ORCA program system. *Wiley Interdiscip. Rev. Comput. Mol. Sci.* 2, 73–78.
- Pamunuwa, G., Karunaratne, D.N., Waisundara, V.Y., 2016. Antidiabetic properties, bioactive constituents, and other therapeutic effects of *Scoparia dulcis*. *Evid. Based. Complement Alternat. Med.* 2016, 8243215.
- Ruiz-Vela, A., Opferman, J.T., Cheng, E.H., Korsmeyer, S.J., 2005. Proapoptotic BAX and BAK control multiple initiator caspases. *EMBO Rep.* 6 (4), 379–385.
- Sharma, K.R., Adhikari, A., Hafizur, R.M., Hameed, A., Raza, S.A., Kalauni, S.K., Miyazaki, J.I., Choudhary, M.L., 2015. Potent insulin secretagogue from *Scoparia dulcis* Linn of Nepalese origin. *Phytother. Res.* 29 (10), 1672–1675.
- Sharma, R.S., Harrison, D.J., Kisielski, D., Cassidy, D.M., McNeilly, A.D., Gallagher, J. R., Walsh, S.V., Honda, T., McCrimmon, R.J., Dinkova-Kostova, A.T., Ashford, M.L. J., Dillon, J.F., Hayes, J.D., 2017. Experimental nonalcoholic steatohepatitis and liver fibrosis are ameliorated by pharmacologic activation of Nrf2 (NF-E2 p45-related factor 2). *Cell Mol Gastroenterol Hepatol* 5 (3), 367–398.
- Shehata, A.M., Quintanilla-Fend, L., Bettio, S., Singh, C.B., Ammon, H.P.T., 2011. Prevention of multiple low-dose streptozotocin (MLD-STZ) diabetes in mice by an extract from gum resin of *Boswellia serrata* (BE). *Phytomedicine* 18 (12), 1037–1044.
- Stephens, P.J., Harada, N., 2010. ECD cotton effect approximated by the Gaussian curve and other methods. *Chirality* 22 (2), 229–233.
- Sun, W.Y., Zhang, Y., Li, Y.Y., Chen, L.Z., Zhang, L., Zhang, C.Y., Zhang, X.P., 2020. Two new labdane diterpenoids from *Scoparia dulcis* to attenuate palmitate-induced viability in MIN6 cells. *Phytochem. Lett.* 38, 25–27.
- Sun, W.Y., Dong, L., Li, Y.Y., Han, L.F., Chen, L.Z., Wang, R.Q., Zhang, L., Zhang, Y., Zhang, X.P., 2021. Progress of research on diterpenoids from *Scoparia dulcis* L. *Progr. Pharmacut. Sci.* 45 (11), 862–867.
- Sybyl Software, 2013. Version X 2.0. Tripos Associates Inc., St. Louis, MO, USA.
- Wan, W.T., Ma, Y.Y., Xu, L.J., Xiao, P.G., 2015. An overview on modern research and application potency of *Scoparia dulcis*. *Chin. Tradit. Herb. Drugs* 46 (16), 2492–2498.
- Wang, H., Huang, M.Y., Bei, W.J., Yang, Y.Q., Song, L.X., Zhang, D.X., Zhan, W.J., Zhang, Y.Z., Chen, X., Wang, W.X., Wang, L.X., Guo, J., 2012. FTZ attenuates liver steatosis and fibrosis in the minipigs with type 2 diabetes by regulating the AMPK signaling pathway. *Biomed. Pharmacother.* 138, 111532.
- Wnęk, A., Andrzejewska, E., Kobos, J., Taran, K., Przewratil, P., 2017. Molecular and immunohistochemical expression of apoptotic proteins Bax, Bcl-2 and Caspase-3 in infantile hemangioma tissues as an effect of propranolol treatment. *Immunol. Lett.* 185, 27–31.
- Xia, M., Shao, J., Qiao, M.M., Luo, Z.G., Deng, X.Z., Ke, Q., Dong, X.X., Shen, L., 2020. Identification of LCA-binding glycans as a novel biomarker for esophageal cancer metastasis using a lectin array-based strategy. *J. Cancer* 11 (16), 4736–4745.
- You, S.F., Zheng, J.Y., Chen, Y.P., Huang, H.B., 2022. Research progress on the mechanism of beta-cell apoptosis in type 2 diabetes mellitus. *Front. Endocrinol.* 13, 976465.
- Zhang, Y., Wang, C., Yu, B., Jiang, J.D., Kong, W.J., 2018. Gastrodin protects against ethanol-induced liver injury and apoptosis in HepG2 cells and animal models of alcoholic liver disease. *Biol. Pharm. Bull.* 41 (5), 670–679.
- Zhang, C.Y., Chen, L.Z., Li, Y.Y., Wei, N., Zhang, L., Dong, L., Zhang, Y., Zhang, X.P., 2020. Scoparic acid E: a new labdane diterpenoid on attenuating palmitate induced viability in MIN6 cells from *Scoparia dulcis*. *Record Nat. Prod.* 14, 383–386.
- Zhang, Y., Wang, R.Q., Yang, Y.N., Ma, N., Zhou, Z., Tan, Y.F., Dong, L., Li, Y.Y., Lu, W. Y., Wu, C.M., Zhang, X.P., 2022. Laurolisins ameliorates type 2 diabetes by regulating the hepatic LKB1-AMPK pathway and gut microbiota. *Phytomedicine* 106, 154423.
- Zhao, Y., Zhang, X.L., Chen, J.Z., Lin, C., Shao, R.F., Yan, C.X., Chen, C., 2016. Hexarelin protects rodent pancreatic β -cells function from cytotoxic effects of streptozotocin involving mitochondrial signalling pathways *in vivo* and *in vitro*. *PLoS One* 11 (2), e0149730.
- Zheng, J.H., Viacava Follis, A., Kriwacki, R.W., Moldoveanu, T., 2016. Discoveries and controversies in BCL-2 protein-mediated apoptosis. *FEBS J.* 283 (14), 2690–2700.